

Generalised Domination and Independence in Graphs

Wayne Goddard *

Department of Computer Science
University of Natal
Durban, 4041 South Africa

Michael A. Henning

Department of Mathematics
University of Natal
Pietermaritzburg, 3209 South Africa

Abstract

In this paper, we introduce a new scheme of parameters which generalises many well-known parameters, including the domination number, the total domination number, the independence number, and the vertex covering number.

1 Introduction

In this paper, we follow the notation of [1]. Specifically, let $G = (V, E)$ be a graph with vertex set V of order n and edge set E , and let v be a vertex in V . The open neighbourhood of v is $N(v) = \{u \in V \mid uv \in E\}$ and the closed neighbourhood of v is $N[v] = \{v\} \cup N(v)$. For a set S of vertices, the open neighborhood of S is defined by $N(S) = \cup_{v \in S} N(v)$, and the closed neighborhood of S by $N[S] = N(S) \cup S$. The subgraph of G induced by the vertices in S is denoted by $\langle S \rangle$.

A set $S \subseteq V$ is a *dominating set* of G if every vertex in $V - S$ is adjacent to a vertex in S . (That is, $N[S] = V$.) The (lower) domination number of G , denoted by $\gamma(G)$, is the minimum cardinality of a dominating set. G . A set $S \subseteq V$ is a *total dominating set* of G if every vertex is adjacent to a vertex in S (other than itself). (That is, $N(S) = V$.) The (lower) total domination number of G , denoted by $\gamma_t(G)$, is the minimum cardinality of a total dominating set.

Two vertices that are not adjacent are said to be independent. A set $S \subseteq V$ is an *independent set* if every two vertices of S are independent. The (upper) *independence*

*Research of both authors supported in part by the University of Natal and the South African Foundation for Research Development.

number of G , denoted by $\beta(G)$, is the maximum cardinality among the independent sets of vertices of G . The set S is called an *independent dominating set* of G if S is both an independent and a dominating set of G . The *independent domination number* $i(G)$ of G is the minimum cardinality among all independent dominating sets of G . Equivalently, $i(G)$ is the minimum cardinality of a maximal independent set of vertices of G .

The concept of domination in graphs, with its many variations, is now well studied in graph theory. The book by Chartrand and Lesniak [1] includes a chapter on domination. For a more thorough study of domination in graphs, see the books edited by Haynes, Hedetniemi and Slater [4, 5].

In this paper, we introduce a new scheme of parameters which generalises many well-known parameters, including the domination number, the total domination number, and the independence number.

2 The Scheme

Let $G = (V, E)$ be a graph. For a set $S \subseteq V$, we define

- $I(S) = S - N(S)$,
- $A(S) = S \cap N(S)$,
- $\partial(S) = N[S] - S$, and
- $Z(S) = V(G) - N[S]$.

For $S \subseteq V$, and for parameters i, a, d, z , we define

$$f(S; i, a, d, z) = i \cdot |I(S)| + a \cdot |A(S)| + d \cdot |\partial(S)| + z \cdot |Z(S)|,$$

and

$$f(G; i, a, d, z) = \max_{S \subseteq V} f(S; i, a, d, z).$$

We call any set S for which this maximum is attained a *realising set* (for $f(G; i, a, d, z)$).

For each set $S \subseteq V$, $f(S; i, a, d, z)$ may be thought of as the profit generated from S where for each element of $I(S)$, $A(S)$, $\partial(S)$ and $Z(S)$ there is a gain of i , a , d and z dollars, respectively. With this interpretation, $f(G; i, a, d, z)$ is the maximum profit over all possible choices for S .

One could also consider the minimum profit, but of course

$$\min_{S \subseteq V} f(S; i, a, d, z) = \max_{S \subseteq V} f(S; -i, -a, -d, -z).$$

3 Examples

We next present examples of parameters that belong to our scheme.

3.1 Domination parameters

We begin with the domination number.

Proposition 1 *For every graph G , $f(G; -1, -1, 0, -\infty) = -\gamma(G)$.*

Proof. Let S be a minimum dominating set of G . Then $Z(S) = \emptyset$. Thus,

$$f(G; -1, -1, 0, -\infty) \geq f(S; -1, -1, 0, -\infty) = -|I(S)| - |A(S)| = -|S| = -\gamma(G).$$

On the other hand, let S be a realising set for $f(G; -1, -1, 0, -\infty)$. Since

$$f(V; -1, -1, 0, -\infty) = -n,$$

we know that $f(S; -1, -1, 0, -\infty) > -\infty$. However,

$$\begin{aligned} f(S; -1, -1, 0, -\infty) > -\infty &\Leftrightarrow Z(S) = \emptyset \\ &\Leftrightarrow S \text{ is a dominating set} \\ &\Leftrightarrow f(S; -1, -1, 0, -\infty) = -|S|. \end{aligned}$$

Thus, $f(G; -1, -1, 0, -\infty) = f(S; -1, -1, 0, -\infty) = -|S| \leq -\gamma(G)$. $\square$

The following results on the independent domination and total domination numbers are proved in a similar fashion to Proposition 1, and so we omit the proofs.

Proposition 2 *For every graph G , $f(G; -1, -\infty, 0, -\infty) = -i(G)$.*

Proposition 3 *For every graph G , $f(G; -\infty, -1, 0, -\infty) = -\gamma_t(G)$.*

These examples illustrate the role of $-\infty$. Suppose S is a realising set for $f(G; i, a, d, z)$. If $i = -\infty$, then the subgraph induced by S cannot contain any isolated vertices, i.e., $I(S) = \emptyset$. If $a = -\infty$, then S must be an independent set, i.e., $A(S) = \emptyset$. If $z = -\infty$, then S must be a dominating set. Hence if both $a = -\infty$ and $z = -\infty$, then S must be a maximal independent set.

3.2 Independence and vertex-cover numbers

It is not surprising that the independence number is given by $\beta(G) = f(G; 1, -\infty, 0, 0)$, but actually the situation is more general.

Proposition 4 *For all $z \leq 0$ and for all graphs G , $f(G; 1, -\infty, 0, z) = \beta(G)$.*

Proof. Let S be a maximum independent set of G . Then $A(S) = Z(S) = \emptyset$. Thus,

$$f(G; 1, -\infty, 0, z) \geq f(S; 1, -\infty, 0, z) = |I(S)| = |S| = \beta(G).$$

On the other hand, let S be a realising set for $f(G; 1, -\infty, 0, z)$. Since $f(G; 1, -\infty, 0, z) \geq \beta(G)$, we know that $f(S; 1, -\infty, 0, z) > -\infty$. However,

$$\begin{aligned} f(S; 1, -\infty, 0, z) > -\infty &\Leftrightarrow A(S) = \emptyset \\ &\Leftrightarrow S \text{ is an independent set.} \end{aligned}$$

We show that S is a maximal independent set. If this is not the case, then $Z(S) \neq \emptyset$. Let $v \in Z(S)$ and let $S' = S \cup \{v\}$. Since S is independent, so too is S' . Thus $A(S') = \emptyset$ and $Z(S') \subseteq Z(S) - \{v\}$. Hence, since $z \leq 0$,

$$\begin{aligned} f(S'; 1, -\infty, 0, z) &= |I(S')| + z \cdot |Z(S')| \\ &\geq (|I(S)| + 1) + z \cdot (|Z(S)| - 1) \\ &= |I(S)| + z \cdot |Z(S)| + 1 - z \\ &> f(S; 1, -\infty, 0, z), \end{aligned}$$

which is a contradiction. Hence S must be a maximal independent set. Thus, $A(S) = Z(S) = \emptyset$, whence $f(G; 1, -\infty, 0, z) = f(S; 1, -\infty, 0, z) = |I(S)| = |S| \leq \beta(G)$. $\square$

Another parameter in the scheme is the vertex cover number. A vertex and an edge are said to *cover* each other in a graph G if they are incident in G . A *vertex cover* in G is a set of vertices that covers all the edges of G . The minimum cardinality of a vertex cover in G is called the *vertex covering number* of G and is denoted by $\alpha(G)$. Equivalently,

$$\alpha(G) = \min |N(S)|,$$

where the minimum is taken over all sets $S \subseteq V$ for which $S = V - N(S)$ (so S is a maximal independent set of G).

Proposition 5 *For all graphs $G = (V, E)$, $f(G; 0, -\infty, -1, -\infty) = -\alpha(G)$.*

Proof. Let $S \subseteq V$ be a set satisfying $S = V - N(S)$ and $\alpha(G) = |N(S)|$. Then S is a maximal independent set of G , so $A(S) = Z(S) = \emptyset$. In particular, $\partial(S) = N(S)$. Thus,

$$f(G; 0, -\infty, -1, -\infty) \geq f(S; 0, -\infty, -1, -\infty) = -|\partial(S)| = -|N(S)| = -\alpha(G).$$

On the other hand, let S be a set satisfying $f(G; 0, -\infty, -1, -\infty) = f(S; 0, -\infty, -1, -\infty)$. Since $f(G; 0, -\infty, -1, -\infty) \geq -\alpha(G)$, we know that $f(S; 0, -\infty, -1, -\infty) > -\infty$. However,

$$\begin{aligned} f(S; 0, -\infty, -1, -\infty) > -\infty &\Leftrightarrow A(S) = Z(S) = \emptyset \\ &\Leftrightarrow S \text{ is a maximal independent set} \\ &\Leftrightarrow f(S; 0, -\infty, -1, -\infty) = -|N(S)|. \end{aligned}$$

Thus, $f(G; 0, -\infty, -1, -\infty) = -|N(S)| \leq -\alpha(G)$. $\square$

3.3 Other members

There are many other members of our scheme that may prove to be of interest. For a graph $G = (V, E)$, we define

- $\nu(G) = \max\{|\partial(S)| : S \subseteq V\}$,
- $i\nu(G) = \max\{|\partial(S)| : S \subseteq V, S \text{ independent}\}$,
- $\eta(G) = \max\{|\partial(S)| - |S| : S \subseteq V\}$, and
- $i\eta(G) = \max\{|\partial(S)| - |S| : S \subseteq V, S \text{ independent}\}$.

These have been studied by Hedetniemi [6] and others. For example, the parameter $\nu(G)$, described in [7], is called the *enclaveless number*. One may show that each of the above parameters is a member of our scheme. The results, the proofs of which we omit, are summarized in Table 1 below.

i	a	d	z	$f(G; i, a, d, z)$
-1	-1	0	$-\infty$	$-\gamma(G)$
-1	$-\infty$	0	$-\infty$	$-i(G)$
1	$-\infty$	0	≤ 0	$\beta(G)$
$-\infty$	-1	0	$-\infty$	$-\gamma_t(G)$
0	$-\infty$	-1	$-\infty$	$-\alpha(G)$
0	0	1	≤ 0	$\nu(G)$
0	$-\infty$	1	≤ 0	$i\nu(G)$
-1	-1	1	0	$\eta(G)$
-1	$-\infty$	1	0	$i\eta(G)$

Table 1

4 Relationships

In this section, we investigate relationships between parameters in our scheme.

4.1 Inequalities

Let G be a graph. We observe firstly that if $i_1 \geq i_2$, then $f(G; i_1, a, d, z) \geq f(G; i_2, a, d, z)$. If $a_1 \geq a_2$, then $f(G; i, a_1, d, z) \geq f(G; i, a_2, d, z)$, and so on. Thus, for example, it follows from Propositions 1 and 2 that $\gamma(G) \leq i(G)$.

4.2 Monotonicity

Proposition 6 *If $a \geq i$ and $d \geq z$, then for all graphs G and for any edge e in the complement of G ,*

$$f(G + e; i, a, d, z) \geq f(G; i, a, d, z),$$

and conversely if $a \leq i$ and $d \leq z$.

Proof. Let $S \subseteq V$ be a realising set for $f(G; i, a, d, z)$. If e joins two vertices of $A(S)$, or of $\partial(S)$, or of $Z(S)$, or if e joins a vertex of $A(S)$ and of $\partial(S)$, then $f(S; i, a, d, z)$ has the same value in $G + e$ as in G , whence $f(G + e; i, a, d, z) \geq f(G; i, a, d, z)$. If e joins two vertices of S , at least one of which belongs to $I(S)$, then adding e to G causes a vertex to move from $I(S)$ to $A(S)$. Since $a \geq i$, this implies that the value of $f(S; i, a, d, z)$ in $G + e$ is at least as large as that in G , whence $f(G + e; i, a, d, z) \geq f(G; i, a, d, z)$. If e joins a vertex of S and a vertex of Z , then adding e to G causes a vertex to move from $Z(S)$ to $\partial(S)$. Since $d \geq z$, this implies that the value of $f(S; i, a, d, z)$ in $G + e$ is at least as large as that in G , and once again $f(G + e; i, a, d, z) \geq f(G; i, a, d, z)$. $\square$

For example, if G is a graph and e an edge in the complement of G , then $\gamma(G + e) \leq \gamma(G)$, by Proposition 1.

4.3 Calculations

We observe the value of $f(G; i, a, d, z)$ when G is a complete graph or a complete bipartite graph.

Proposition 7 *For all integers $n \geq 2$,*

$$f(K_n; i, a, d, z) = \max\{an, nz, i + (n - 1)d, a + (n - 2)d\}.$$

Proof. Let $S \subseteq V$ and $|S| = s$. We claim one need only consider the four cases $s = 0, 1, 2, \infty$. For suppose $2 \leq s \leq n$. Then $I(S) = Z(S) = \emptyset$ so that $f(S) = as + (n - s)d = nd + (a - d)s$. This function is linear in s and is maximised either at $s = 2$ or $s = n$. $\square$

Proposition 8 *For all integers $m \geq 2$,*

$$f(K_{m,m}; i, a, d, z) = \max\{ na, nz, m(i+d), i+md+(m-1)z, 2a+(n-2)d \},$$

where $n = 2m$.

Proof. Let $S \subseteq V$, and let $r = |S \cap V_1|$ and $t = |S \cap V_2|$ where $K_{m,m}$ has partite sets V_1 and V_2 . We may assume that $r \geq t$. We claim that one need only consider the five cases $(r, t) = (0, 0), (1, 0), (m, 0), (1, 1)$ and (m, m) .

For suppose first that $r > 0$ and $t = 0$. Then $f(S) = ir + z(m-r) + dm$ which as a function of r is maximised at one of the extreme values $r = 1$ or $r = m$. Suppose second that $r, t > 0$. Then $f(S) = a(r+t) + d(2m-r-t)$ which as a function of $r+t$ is maximised at one of the extreme values $r+t = 2$ or $r+t = 2m$. $\square$

4.4 Gallai-type results

Let G be a graph of order n . A Gallai-type result is a result of the type $\psi(G) + \phi(G) = n$ for some graph-theoretical parameters ψ and ϕ . For a recent paper on Gallai-type results, see for example [8].

For $S \subseteq V$, and for constants b and c , it follows from the definition of our scheme that

$$f(S; bi+c, ba+c, bd+c, bz+c) = b \cdot f(S; i, a, d, z) + cn.$$

Hence we get the following Gallai-type result for our scheme.

Proposition 9 *For all graphs G of order n ,*

$$f(G; i, a, d, z) - f(G; i-1, a-1, d-1, z-1) = n.$$

As a consequence follow the well-known results that the independence number is the "Gallai-dual" of the vertex covering number (see [3]) and the enlaveless number is the "Gallai-dual" of the domination number (see [7]).

Corollary 10 *For all graphs G of order n*

- (a) $\alpha(G) + \beta(G) = n$.
- (b) $\gamma(G) + \nu(G) = n$.

Proof. (a) By Proposition 4 with $z = -\infty$, we know that $\beta(G) = f(G; 1, -\infty, 0, -\infty)$ and by Proposition 5, we know that $\alpha(G) = -f(G; 0, -\infty, -1, -\infty)$.

(b) From Table 1 with $z = -\infty$, we know that $\nu(G) = f(G; 0, 0, 1, -\infty)$ and by Proposition 1, we know that $\gamma(G) = -f(G; -1, -1, 0, -\infty)$. $\square$

In fact, Gallai "duals" are automatic for members of the scheme. For example, the function $-f(G; -2, -2, 0, -1)$ provides the Gallai dual for $\eta(G) = f(G; -1, -1, 1, 0)$.

4.5 Maximum and minimum values

Consider the four values i, a, d, z . If z is maximum, then every graph G has $f(G; i, a, d, z) = nz$. If a is maximum, then every graph G without isolates has $f(G; i, a, d, z) = na$. The interesting cases are where i or d is the maximum value.

Proposition 11 *For fixed values of i, a, d, z and of the order n of graph G , $f(G; i, a, d, z)$ is maximised for G empty or for G a star (though the extremal graphs might not be unique).*

Proof. If z is maximum, then any graph is extremal. If a is maximum, then any connected graph is extremal. If i is maximum, then the empty graph is extremal.

Suppose d is maximum. Then if $\partial(S) \neq \emptyset$, we must have $|I(S)| \geq 1$ or $|A(S)| \geq 2$. On the star we can have $|I(S)| = 1$ and $|\partial(S)| = n - 1$, and can have $|A(S)| = 2$ and $|\partial(S)| = n - 2$, so that the star is extremal. $\square$

If one insists that the graph be connected, then the star is always maximum.

Proposition 12 *For fixed values of i, a, d, z and of the order n of graph G , $f(G; i, a, d, z)$ is minimised for G empty or for G complete (though the extremal graphs might not be unique).*

Proof. If z is maximum, then any graph is extremal. If a is maximum, then the empty graph is extremal.

Suppose i is maximum. If $z \geq d$, then by Proposition 6 the complete graph is extremal. So we may assume that $d > z$. If S is a maximal independent set, then $A(S) = Z(S) = \emptyset$ and so $f(S; i, a, d, z) \geq |S|i + (n - |S|)d \geq i + (n - 1)d$. Also, $f(V; i, a, d, z) \geq na$. Thus

$$f(G; i, a, d, z) \geq \max\{i + (n - 1)d, na\}.$$

But this value is achieved for $G = K_n$.

Suppose d is maximum. If S is a maximal independent set, then $A(S) = Z(S) = \emptyset$ and so $f(S; i, a, d, z) \geq |S|i + (n - |S|)d \geq ni$. Also, $f(\emptyset; i, a, d, z) = nz$. Thus

$$f(G; i, a, d, z) \geq \max\{ni, nz\}.$$

But this value is achieved for $G = nK_1$. $\square$

If we insist that the graph be connected, then the cases where a, i or z are maximum are easily handled. The case where d is maximum is interesting and we do not have a complete characterisation. For example, the enclaveless number $\nu(G) = f(G; 0, 0, 1, 0)$ of G is minimised for the graph formed by taking a clique on $n/2$ vertices and an independent set on $n/2$ vertices and adding a matching between them.

5 Algorithms

Some of the parameters in the scheme are known to be NP-complete (for example, domination number) while others have polynomial-time algorithms (for example, $f(1, 1, 0, 0) = n$). It might be nice to prove a general result.

5.1 A Linear Algorithm on Trees

There is a linear-time algorithm for finding $f(T; i, a, d, z)$ for a tree T for any fixed value of i, a, d and z . It reduces to known algorithms for parameters such as domination number and independence number (see, for example, [2]).

The algorithm roots the tree T and associates various variables with the vertices of T as it proceeds starting with the vertices at highest levels and finishing with the root. More precisely, for a rooted tree $T_v = (V, E)$ with root v , define

$$f_C(T_v) = \max_{S \subseteq V} f(S; i, a, d, z),$$

subject to the condition that $v \in C(S)$ where $C(S) \in \{I(S), A(S), \partial(S), Z(S)\}$. We then associate with the root v the quartet $f(T_v) = (f_A(T_v), f_I(T_v), f_\partial(T_v), f_Z(T_v))$. The value of $f(T_v)$ can be calculated by a postorder traversal i.e., by calculating the values for the subtree rooted at the children and then the value for the whole tree.

For example, suppose the root v has children set W . In order to calculate $f_A(T_v)$, the only restriction on S is that the root v must be in S and at least one of the children must be in S . Hence the formula for $f_A(T_v)$ is obtained by calculating the best we can do with total freedom in the children and then subtracting the penalty for the restriction that at least one child is in S . Thus, letting

$$g(T_v) = \sum_{w \in W} \max(f_A(T_w), f_I(T_w) + a - i, f_\partial(T_w), f_Z(T_w) + d - z),$$

and

$$h(T_v) = \min_{w \in W} (\max(f_\partial(T_w), f_Z(T_w) + d - z) - \max(f_A(T_w), f_I(T_w) + a - i)),$$

we have

$$f_A(T_v) = a + g(T_v) - \max(0, h(T_v)).$$

In order to calculate $f_I(T_v)$, the only restriction is that the root v must be in S and no child of v must be in S . Hence the formula for $f_I(T_v)$ is obtained by calculating the best we can do with no child of v in S . Thus,

$$f_I(T_v) = i + \sum_{w \in W} \max(f_\partial(T_w), f_Z(T_w) + d - z).$$

In order to calculate $f_{\partial}(T_v)$, the only restriction is that the root v must not be in S and at least one child of v must be in S . Hence the formula for $f_{\partial}(T_v)$ is obtained by calculating the best we can do with total freedom in the children and then subtracting the penalty for the restriction that at least one child is in S . Thus, letting

$$\ell(T_v) = \sum_{w \in W} \max(f_A(T_w), f_I(T_w), f_{\partial}(T_w), f_Z(T_w)),$$

and

$$k(T_v) = \min_{w \in W} (\max(f_{\partial}(T_w), f_Z(T_w)) - \max(f_A(T_w), f_I(T_w))),$$

we have

$$f_{\partial}(T_v) = d + \ell(T_v) - \max(0, k(T_v)).$$

Finally, in order to calculate $f_Z(T_v)$, the only restriction is that the root v must not be in S and no child of v must be in S . Hence the formula for $f_Z(T_v)$ is obtained by calculating the best we can do with no child of v in S . Thus,

$$f_I(T_v) = z + \sum_{w \in W} \max(f_{\partial}(T_w), f_Z(T_w)).$$

This can be calculated in time proportional to $|W|$.

References

- [1] G. Chartrand and L. Lesniak, *Graphs & Digraphs: Third Edition*, Chapman & Hall, London, 1996.
- [2] E.J. Cockayne, S.E. Goodman, and S.T. Hedetniemi, A linear algorithm for the domination number of a tree. *Inform. Process. Lett.* **4** (1975), 41–44.
- [3] T. Gallai, Über extreme Punkt- und Kantenmengen. *Ann. Univ. Sci. Budapest, Eötvös Sect. Math.* **2** (1959), 133–138.
- [4] T.W. Haynes, S.T. Hedetniemi, and P.J. Slater, *Fundamentals of Domination in Graphs*, Marcel Dekker, New York. To appear in 1997.
- [5] T.W. Haynes, S.T. Hedetniemi, and P.J. Slater (eds), *Domination in Graphs : Advanced Topics*, Marcel Dekker, New York. To appear in 1997.
- [6] S.T. Hedetniemi, Seminar talk, Clemson, 1994.
- [7] P.J. Slater, Enclaveless sets and MK-systems, *J. of Research of the National Bureau of Standards* **82** (1977), 197–202.
- [8] P.J. Slater, Generalized graph parameters: Gallai Theorems I, *Bulletin of the ICA* **17** (1996), 27–37.